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Electronic Circuits

Fundamentals and Applications

Mike Tooley



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Electronic Circuits: Fundamentals and Applications

Third Edition

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Preface

This is the book that I wish I had when I first started exploring electronics nearly half a century ago. In those days, transistors were only just making their debut and integrated circuits were completely unknown. Of course, since then much has changed but, despite all of the changes, the world of electronics remains a fascinating one. And, unlike most other advanced technological disciplines, electronics is still something that you can ‘do’ at home with limited resources and with a minimal outlay. A soldering iron, a multi-meter, and a handful of components are all that you need to get started. Except, of course, for some ideas to get you started—and that’s exactly where this book comes in!

The book has been designed to help you understand how electronic circuits work. It will provide you with the basic underpinning knowledge necessary to appreciate the operation of a wide range of electronic circuits including amplifiers, logic circuits, power supplies and oscillators.

The book is ideal for people who are studying electronics for the first time *at any level* including a wide range of school and college courses. It is equally well suited to those who may be returning to study or who may be studying independently as well as those who may need a quick *refresher*. The book has 19 chapters, each dealing with a particular topic, and eight appendices containing useful information. The approach is topic-based rather than syllabus-based and each major topic looks at a particular application of electronics. The relevant theory is introduced on a progressive basis and delivered in manageable chunks.

In order to give you an appreciation of the solution of simple numerical problems related to the operation of basic circuits, worked examples have been liberally included within the text. In addition, a number of problems can be found at the end of each chapter and solutions are provided at the end of the book. You can use these end-of-chapter problems to check your understanding and

also to give you some experience of the ‘short answer’ questions used in most in-course assessments. For good measure, we have included 70 revision problems in Appendix 2. At the end of the book you will find 21 sample coursework assignments. These should give you plenty of ‘food for thought’ as well as offering you some scope for further experimentation. It is not envisaged that you should complete all of these assignments and a carefully chosen selection will normally suffice. If you are following a formal course, your teacher or lecturer will explain how these should be tackled and how they can contribute to your course assessment. While the book assumes no previous knowledge of electronics you need to be able to manipulate basic formulae and understand some simple trigonometry in order to follow the numerical examples. A study of mathematics to GCSE level (or equivalent) will normally be adequate to satisfy this requirement. However, for those who may need a refresher or have had previous problems with mathematics, Appendix 6 will provide you with the underpinning mathematical knowledge required.

In the later chapters of the book, a number of representative circuits (with component values) have been included together with sufficient information to allow you to adapt and modify the circuits for your own use. These circuits can be used to form the basis of your own practical investigations or they can be combined together in more complex circuits.

Finally, you can learn a great deal from building, testing and modifying simple circuits. To do this you will need access to a few basic tools and some minimal test equipment. Your first purchase should be a simple multi-range meter, either digital or analogue. This instrument will allow you to measure the voltages and currents present so that you can compare them with the predicted values. If you are attending a formal course of instruction and have access to an electronics laboratory, do make full use of it!